

# JERRY ZACHARY BANNISTER


Statistician; Data Scientist; PhD Researcher in Statistics

**BUSINESS:** Founder, JJMB Datascience Hub; **LOCATION:** Bauchi, Nigeria

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**GOOGLE SCHOLAR:** scholar.google.com/citations?user=oKaejtMAAAAJ


## PROFILE

Statistician, data scientist and business founder with more than ten years of turning incomplete Nigerian data into evidence that institutions act on. Since 2015 I have run the statistical side of the Academic Planning and Quality Assurance Directorate at Abubakar Tafawa Balewa University (ATBU), Bauchi, producing the university's federal returns to the National Universities Commission (NUC) and the analyses that go to Senate. Since 2017 I have run a statistical consultancy, now registered as **JJMB Datascience Hub**, that serves researchers, NGOs and businesses. My research develops uncertainty-aware statistical methods for health, poverty and disability-inclusion data in fragile settings. Co-author of seven peer-reviewed articles; MSc Statistics, best in class (top 1%; GPA 4.40/5.00); PhD in progress.

## DISABILITY AND LIVED EXPERIENCE

I am a researcher and a wheelchair user. I have faced physical barriers and social discrimination, and at other times favouritism, as people singled me out because of my disability. Both taught me the same thing: disabled people need equal access, not exceptions. That lived experience shapes what I build. My disability-inclusion tools are accessible by design (screen-reader-safe layouts, voice input and output, keyboard navigation, high contrast) because I know first-hand what inaccessible buildings, services and systems cost disabled people.

**Proposed fellowship and grant focus:** open, uncertainty-aware evidence on disability and poverty in Nigeria. The work would link state-level poverty and health data with disability indicators, and develop my Nigeria Equity Atlas, Disability Rights Assistant and Impact Mosaic prototypes into validated, accessible tools for disabled people's organisations and policymakers.

## AREAS OF CONTRIBUTION

- **Disability inclusion and population health:** Bayesian spatial and survival models, poverty and health atlases, and accessible decision tools.
- **Institutional data and quality assurance:** ten years of federal regulatory reporting, accreditation evidence, teaching-quality evaluation and academic-integrity analysis for a federal university.
- **Data science entrepreneurship:** consultancy, training and deployed software for researchers, NGOs and businesses on low-cost, reproducible infrastructure.

## EDUCATION

### PhD Statistics (in progress)

2026 – present

*Abubakar Tafawa Balewa University (ATBU), Bauchi*

Thesis: A Hybrid Bayesian Deep Learning Framework for Non-Stationary Spatio-Temporal Terrorism Risk Forecasting and Uncertainty-Aware Safety Decision Support in Nigeria. Seminar proposal stage.

### MSc Statistics; Best in Class (top 1%), GPA 4.40 / 5.00

2024

*Abubakar Tafawa Balewa University (ATBU), Bauchi; 2022/2023 session; awarded with effect from 14 May 2024*

Thesis: Close-Knit Regression as a single imputation technique for data missing completely at random (supervisor: Prof. A. Abdulkadir). Published (publication 5).

**BSc (Hons) Statistics, Second Class Upper; Second Best in Class (top 2%), GPA 4.20 / 5.00** 2008 – 2012

*University of Maiduguri (UNIMAID), Borno; certificate issued 15 April 2014*

Project: M/M/c queuing analysis of automated teller machines, Fidelity Bank, Maiduguri.

**NECO Senior School Certificate (SSCE), June 2008: 7 credits** 2002 – 2008

*International Secondary School, ATBU, Bauchi*

**First School Leaving Certificate** 2000 – 2002

*Total Child School, Bauchi*

## PROFESSIONAL EXPERIENCE

**Abubakar Tafawa Balewa University, Bauchi** 2014 – present

*Directorate of Academic Planning and Quality Assurance (AP&QA)*

Career progression: National Youth Service (NYSC), AP&QA, 2014–2015; Statistician II, 2015–2021; Statistician I, 2021–2024; Senior Statistician, 2024–present.

### *Core duties*

- Compile, validate and report the university's student and staff statistics to the NUC and other federal bodies, including the data behind NUC annual review meetings.
- Prepare evidence for programme accreditation and for Senate review of new programmes against NUC Minimum Academic Standards.
- Built a Python/SQLite desktop application that manages the university's 223 professor records and generates the NUC update returns, replacing manual compilation.
- Design and analyse institutional surveys for quality control and programme evaluation; produce data briefs and presentations for university management.

### *Selected institutional analyses*

- **Academic misconduct analysis for the Senate Business Committee (2025).** Analysed the confirmed examination-misconduct cases for the first semesters of 2022/23 and 2024/25, adjusted for enrolment. The rate rose from 0.93 to 4.2 cases per 1,000 students. Carry-over students accounted for 9% of cases in 2022/23 and 32% in 2024/25. The analysis also identified the high-risk venues, offence types and faculties. Presented to the Senate Business Committee by the Acting Director.
- **Appraisal of classroom facilities, all faculties (2nd semester 2015/16).** Built the indices for capacity utilisation, facility condition, functional carrying capacity and timetable–venue fit. Overall hall utilisation was 80%, against a Nigerian standard of 51%, with marked under-use in some faculties. The results informed infrastructure planning.
- **Evaluation of shared courses across all faculties (2014/15, reported 2016).** Analysed grade distributions and performance indices from departmental score sheets for courses taken by students of several departments, as the evidence base for targeted teaching interventions.
- **Performance rating of courses (1st semester 2017/18).** Analysed the course-rating survey data and produced the tables, charts, faculty reports and Directorate presentation.
- **Lecturer performance and classroom facilities evaluation (AP&QA–SERVICOM).** Statistician and data scientist for the joint evaluation of lecturer organisation and preparation, teaching skills and the quality of classroom facilities across the university.
- **Undergraduate enrolment trends, 2011–2022.** Faculty-level growth analysis for the Director's records, e.g. Engineering enrolment up 105.5% and Environmental Technology up 74.6% over the period.

### *Institutional data systems built for the Directorate (internal to ATBU; verifiable on request)*

*These systems hold confidential student and staff records, so they are not public. ATBU can confirm them on request.*

- **NUC statistical digest automation and validation engine.** A scripted pipeline that combines departmental returns into the annual NUC digest (Forms 1–11) and checks cross-form totals, missing entries and out-of-range values before submission.

- **University Factbook 2021–2025 data pipeline.** A pipeline that rebuilds the Factbook from the raw yearly returns, so every table and chart comes from code and is traced to its source file.
- **Enrolment and registration trend monitor.** Combines enrolment and registration data from 2012/13 to 2024/25 (by programme, state and sex) into one panel and flags structural breaks and implausible reporting jumps for management briefings.
- **Lecturer performance and course-rating analytics platform.** Measures the reliability of each rating instrument, compares faculties and departments with multilevel models, flags courses on control charts and generates the faculty reports.
- **Shared-course performance evaluation pipeline.** Compares grade distributions between departments taking the same course, with effect sizes, Holm-adjusted comparisons and a performance index ranking.
- **Examination misconduct analytics and early-warning dashboard.** Case-level misconduct data with rates per 1,000 students modelled by Poisson regression, tests of association between offence type and venue, and early warnings for high-risk venues, reported to the Senate Business Committee.
- **Academic staffing, promotion-pipeline and benchmark-ratio analytics.** Combines the annual staff workbooks into one validated staffing model, tracks promotion cases by faculty and rank, and monitors NUC benchmark ratios (PhD to MSc, student to staff, teaching to non-teaching).
- **Programme portfolio register and curriculum-overlap analyser.** A register of programmes with their accreditation status, a text-similarity tool that detects duplicated courses across departmental curricula, and Senate evidence packs for proposed programmes.

Received a Directorate commendation letter for resilience and hard work (2021).

**JJMB Datascience Hub, Bauchi** Jan 2017 – present

*Founder and Lead Data Scientist; CAC-registered 2026; trading as Frequently Bayesian Data Analytics since 2017*

- Statistical consultancy, data-skills training and software for postgraduate researchers, faculty, NGOs and businesses in public health, veterinary science, microbiology, agriculture and the social sciences: study design, sample size, modelling, analysis and write-up.
- Built the reproducible screening and evidence-synthesis workflow for a PRISMA 2020 systematic review of avian influenza in Nigeria, 2006–2026 (2,157 records screened; 113 studies included).
- Bayesian, spatial and survival analyses of survey and clinical data, including Nigeria DHS data and hospital TB cohorts.
- Delivers end-to-end products on free cloud infrastructure, from ingestion to deployed dashboards. Example: a price-intelligence pipeline with an XGBoost demand forecast at 10.9% MAPE against a 13.8% naive baseline.
- Reproducible R/Quarto and Python pipelines that take raw laboratory and field spreadsheets to client-ready reports.

## PUBLICATIONS

*Peer-reviewed journal articles (* corresponding author; published as J. B. Zachary or J. Bannister). Full list: ORCID; Google Scholar*

1. Ben, E. O., **Zachary, J. B.**, Mashat, S. Y., & Chiege, N. U. (2025). Modern prediction models for BMI: Contrasting XGBoost with Bayesian geo-additive quantile regression model in women's health. *Asian Journal of Probability and Statistics*, 27(7), 101–113. doi:10.9734/ajpas/2025/v27i7779
2. Magani, J. J., Baba, A. M., Bawa, M. U., Dunga, J. A., Lawan, A. M., & **Zachary, J. B.** (2025). Survival analysis of MDR-TB patients using parametric model: A case study of ATBU Teaching Hospital MDR-TB Centre, Bauchi. *International Journal of Innovative Mathematics, Statistics & Energy Policies*, 13(4), 41–54. doi:10.5281/zenodo.17403267
3. Oguche, M. O., Adams, Y. Y., Apollos, E., Emmanuel, H. S., Emmanuel, L. L., Joshua, M. B., Samson, M., **Zachary, J. B.**, & Okonkwo, I. G. (2025). Epidemiological characteristics and prevalence of gastrointestinal parasites in small ruminants in Toro LGA, Bauchi State, Nigeria. *International Journal of Research and Innovation in Social Science*, 9(10), 5749–5755. doi:10.47772/IJRISS.2025.910000471
4. Yusuf, N., & **Zachary, J. B.** (2023). Threshold effects on outlier detection: A comparative study of MCD and MRCD estimators in multivariate data analysis. *Asian Journal of Probability and Statistics*, 25(2), 117–126. doi:10.9734/ajpas/2023/v25i2557

5. Abdulkadir, A., **Zachary, J. B.***, Yusuf, N., & Musa, K. (2023). Close-Knit-Regression: An efficient technique in estimating missing completely at random data. *Asian Journal of Probability and Statistics*, 24(3), 48–59. doi:10.9734/ajpas/2023/v24i3528
6. Akpensuen, S. H., Hassan, S., & **Bannister, J.** (2022). Robust estimation of GARCH model to appraise excess kurtosis. *World Scientific News*, 167, 69–85.
7. Daniel, S., Lasisi, K. E., & **Bannister, J.** (2020). Application of survival analysis of TB patients using parametric model: A case study of General Hospital Bayara. *Asian Journal of Probability and Statistics*, 6(4), 54–65. doi:10.9734/ajpas/2020/v6i430169

### *Manuscripts and books*

- **Zachary, J. B.** (in preparation). Bayesian change-point detection in 74 years of Nigerian rainfall: coherent but non-simultaneous drying across three climate zones. First author; full draft.
- Avian influenza in Nigeria, 2006–2026: a PRISMA 2020 systematic review. Manuscript complete; evidence-synthesis pipeline built by me.
- **Zachary, J. B.** (2026). Urban agriculture and urban food security in Nigeria: a spatial-statistics and open-data analysis. Manuscript prepared for TeMA - Journal of Land Use, Mobility and Environment, 2026. First author; full draft
- **Zachary, J. B.** (2017). *Mathematics for Junior Secondary Schools.* MEG Publishers, Bauchi. ISBN 978-978-52955-9-7.

## GRANTS AND PROPOSALS

- Research team member, TETFund National Research Fund 2026 concept note (prepared): *Development and validation of an AI-enabled, safe, affordable local-food decision-support tool for complementary feeding in Northeast Nigeria.* 24 months; provisional budget NGN 42 million.

## RESEARCH SOFTWARE AND OPEN SCIENCE

*Selected from 26 public repositories, each with code, data provenance and documentation: github.com/batestguy*

### *Disability inclusion, poverty and health*

- **Curb Cut** (Python, dbt, DuckDB-WASM, JavaScript). Accessible, zero-cost platform for Nigerian disability data, in active development: the first phase is live, the chart library is in progress, and the analysis phases are planned. It extracts JONAPWD's state-by-type figures for all 37 states (370 values) from the source PDF and publishes them beside the source's own caveat that they assume a 15% rate rather than count people. A dbt and DuckDB warehouse builds the public marts (51 tests) and the site queries them in the browser, so it needs no server and publishes only aggregates; the NBS MPI 2021 microdata (213,820 people aged 5 and over, Washington Group Short Set), GHS-Panel and DHS files stay in a git-ignored local zone. Accessibility is tested rather than claimed: nine Playwright and axe tests cover WCAG 2.2 AA in light and dark, keyboard order, accessible names, table semantics and 320 px layouts, and an automated NVDA run confirmed the skip link, heading order, chart description and table headers. Planned next: survey-weighted state prevalence with confidence intervals and small-area estimation, a pre-registered poverty analysis, and chart descriptions in Hausa, Yoruba, Igbo and Pidgin. Human screen-reader sign-off is outstanding.
- **Disability Rights Legal and Connection Assistant** (Python, RAG, Streamlit). Multi-turn assistant on Nigeria's Disability Act 2018 and the 1999 Constitution, with a verified NGO and helpline connector. Accessible by design. Live demo.
- **Impact Mosaic** (React, TypeScript). Monitoring, evaluation and learning prototype for a disability-inclusion programme: role-based views, scorecards, and state- and LGA-level PDF reports. Uses fictional data.
- **Nigeria Equity Atlas** (Python, Tableau). Multidimensional Poverty Index for all 36 states and the FCT (MICS 2021) by health, education and living standards, with conflict and climate overlays and a four-round trend.

- **Dandali Health Facility Dashboard** (JavaScript, Leaflet). Maps 51,022 health facilities live from the GRID3 government data service; works offline.

*Bayesian modelling, evidence synthesis and decision support*

- **Bayesian change-point detection in Nigerian rainfall** (PyMC). Hierarchical models on 74 years of data from nine stations; LOO cross-validation decisively rejects the no-change model in every zone.
- **Avian influenza systematic review pipeline** (Python). Records from six databases deduplicated into one register; 2,157 records screened following PRISMA 2020, PRISMA-S and SWiM.
- **Crude Blend Optimizer** (Python, Docker). ML surrogate models (5-fold $R^2 \geq 0.982$) with a differential-evolution optimiser that comes within $0.005 per barrel of the LP optimum; 140 tests.
- **Also:** Fieldmargin offline crop-margin planner (React); Sol Searching solar lead generation (Streamlit); Bayesian World Cup model with a 50,000-run tournament simulation (PyMC); applied biostatistics pipelines for anti-trypanosomal and *E. coli* studies (R, Quarto).

*Further projects and resources*

- **Causal inference:** interactive explorer of Nigerian presidential transitions and national outcomes, 1999 to the present, with a provenance-verified data-ingestion pipeline (in development).
- **Civic and electoral data:** Bauchi Voter Pulse, a source-backed, bilingual campaign-intelligence website for the 2027 Bauchi governorship, and an offline ward-level campaign command centre covering 20 LGAs and 212 wards on GRID3 geography (vanilla JavaScript, single file).
- **Sports analytics:** Messi vs Ronaldo, with difficulty-weighted goal rates and match-level bootstrap uncertainty (R, Shiny); European vs South American football comparison (Python).
- **Teaching and learning resources:** Quarto blog of statistical and ML case studies and simulations; an Anki deck of 261 cards covering all eight swirl R courses; a 3,214-card chess endgame deck from Silman's *Complete Endgame Course*.
- **Earlier projects:** frequentist and Bayesian tests of a mobile network's night-data tariff delivery (2022); sexuality and natural disasters, a Bayesian assessment (2022); machine-learning feature analysis of Nigerian Spotify charts (2021).

## METHODS AND TECHNICAL SKILLS

- **Statistics:** Bayesian hierarchical models and MCMC; survival analysis; spatial and geo-additive regression; time series and GARCH; robust multivariate methods; missing-data imputation; survey and experimental design; causal inference; evidence synthesis.
- **Machine learning:** scikit-learn, XGBoost, TensorFlow/Keras, transformer fine-tuning, retrieval-augmented generation, optimisation and Monte Carlo simulation.
- **Programming and deployment:** Python, R, SQL, TypeScript; Git/GitHub Actions, Docker, Quarto/R Markdown; Streamlit, Shiny, React; Tableau, Power BI; SPSS, Stata, Minitab.

## AWARDS, TRAINING AND SERVICE

- Best in Class, MSc Statistics, ATBU (2024); Commendation Letter, Directorate of Academic Planning, ATBU (2021); MTN Foundation Scholarship and Dean's List, University of Maiduguri (2009–2012).
- Training: ICDL (2021); Data Science and Machine Learning, OLAP Computers (2020); ICT Advanced Online Training, Mankato Digital Age Centre (2020); Advanced Digital Appreciation Programme for Tertiary Institutions (ADAPTI), ATBU (2016).
- Membership and service: Member, Chartered Institute of Statisticians of Nigeria (CISON); Senior Staff Association of Nigerian Universities (SSANU), ATBU Chapter (2020–); Nigerian Association of Mathematical Sciences Students (NAMSS), UNIMAID Chapter (2012).
- Interests: chess, with a peak rating of 2000+ (rapid, Chess.com; blitz, Lichess).

## REFEREES


### Prof. Ahmed Abdulkadir

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## PERSONAL DETAILS

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|---|---|---|---|
| DATE AND PLACE OF BIRTH | 13 January 1993, Bauchi | NATIONALITY | Nigerian |
| STATE OF ORIGIN | Gombe | SEX | Male |
| DISABILITY | wheelchair user | INSTITUTIONAL EMAIL | jzbannister@atbu.edu.ng |
| PERSONAL EMAIL | batesthommie@gmail.com | PHONE | +234 706 167 4831 |

## PERSONAL STATEMENT

I was born in Bauchi in 1993 and have lived and worked there for most of my life. I am a wheelchair user. Getting around a Nigerian campus, office or city in a wheelchair means meeting physical barriers every day: steps with no ramp, offices on upper floors, roads and transport not built for me. I have also faced social discrimination, people who assume that a disabled person cannot lead, analyse or decide. At other times I have been favoured, given exceptions out of pity rather than on merit. Both responses treat disability as the defining fact about a person. Neither gives what disabled people actually need, which is equal access and fair judgement.

I have built my career on being judged by results. I graduated second best in my BSc Statistics class at the University of Maiduguri, supported by an MTN Foundation scholarship, and best in my MSc Statistics class at Abubakar Tafawa Balewa University (ATBU), Bauchi (top 1%, GPA 4.40/5.00). My MSc thesis became a published imputation method, Close-Knit Regression. I am now a PhD researcher in Statistics, developing Bayesian deep-learning methods that forecast risk with honest uncertainty.

For more than ten years I have worked in ATBU's Directorate of Academic Planning and Quality Assurance, rising from National Youth Service to Senior Statistician. I produce the statistics the university reports to the National Universities Commission and the evidence that Senate uses on accreditation, teaching quality and examination integrity. I have built the Directorate's internal data systems, from NUC digest validation to misconduct analytics and staffing models. Since 2017 I have also run a statistical consultancy, registered in 2026 as JJMB Datascience Hub, that supports researchers, NGOs and businesses, and I have co-authored seven peer-reviewed articles.

My lived experience now drives my research. Disabled Nigerians are close to invisible in the data that governments use to plan: disability is often missing from surveys, poorly measured when it is collected, and rarely linked to poverty and health outcomes. Where the data are thin, decisions are made by assumption, and assumptions are where discrimination lives. I have started to close this gap with open, accessible tools. They include a Disability Rights Assistant that answers questions on the Disability Act 2018 with voice input, keyboard navigation and screen-reader support; Impact Mosaic, a monitoring and evaluation prototype for disability-inclusion programmes; and the Nigeria Equity Atlas of multidimensional poverty across all 36 states and the FCT.

With grant support I will turn these prototypes into validated, sustained resources. I will link state-level poverty and health data with disability indicators using Bayesian methods that report uncertainty honestly, so that policymakers see both what the evidence shows and how confident they can be. I will co-design the tools with disabled people's organisations so that they answer the questions those organisations actually face, and publish all code and methods openly. I will also train other disabled people in data skills through JJMB Datascience Hub, because the most durable way to make disabled people visible in data is for disabled people to produce it.

Funding would let me spend protected time on this work, meet the extra costs that accessible travel and fieldwork impose on a wheelchair user, and reach the organisations who can use the results. I bring ten years of institutional experience, a record of delivering working software and peer-reviewed research, and a first-hand understanding of what exclusion costs. I ask to be judged, as I always have been, on what I deliver.

*Jerry Zachary Bannister · Bauchi, October 2026*